



10523 - Distilling complex organics in the iconic protostellar system IRAS 16293-2422

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
MIRI MRS				
	1	science	MIRI Medium Resolution Spectroscopy	(1) IRAS-16293-2422
	2	background	MIRI Medium Resolution Spectroscopy	(2) bkg

ABSTRACT

Some infant protostars are accompanied by a striking level of chemical complexity in the gas that enshrouds them. Of these, the Class 0 low-mass protostellar system IRAS 16293-2422 is the archetype with a decades-long legacy of observational studies at (sub)millimeter wavelengths. These studies have yielded the most robust characterization of the chemical composition of the gas in this young system, including its complete complex organic gas-phase inventory. However, what is entirely missing is constraints on its complex organic ices. The mid-IR continuum of IRAS 16293-2422 was prohibitively faint for ice studies with past missions such as ISO and Spitzer. However, with the JWST, we can now finally unravel what complex species may be lurking as ices in this iconic star-forming system. We propose MIRI MRS observations to close the gap in our understanding of the chemical complexity in ices towards this source. Our observational campaign will make a major step forward in understanding (1) what icy

materials are available in the innermost regions for the assembly of protoplanetary disks at the earliest Class 0 stage of star formation; and (2) what physicochemical processes shape the chemical complexity of volatiles that are eventually incorporated into planetesimals and comets in a protostellar system that closely resembles our early Solar System.

OBSERVING DESCRIPTION

We propose MIRI MRS observations of the A & B components of the iconic protostellar system IRAS 16293-2422 to characterize the chemical complexity of its ices for comparison with its robustly quantified gas-phase complex organic inventory. With the MIRI IFU, we can spatially disentangle the two components of this system and by executing a 2x2-pointing mosaic, we can capture both of them in one observation. JWST's exquisite sensitivity allows us to easily constrain the spectral energy distributions (SEDs) of the two targets down to ~ 1 mJy at the shortest mid-IR wavelengths

and to search for complex organic spectral features at a SNR of $\sim >5$ in the 7-8.5 micron range. The chemically simple ices in its enshrouding envelope have already been characterized with complimentary Cycle 2 NIRSpec MSA observations (program #3222). MIRI observations of complex organic ices in IRAS 16293-2422 are the last missing puzzle piece in our understanding of this iconic source, which has shaped much of our understanding of interstellar chemistry in low-mass star-forming regions.

Proposal 10523 - Targets - Distilling complex organics in the iconic protostellar system IRAS 16293-2422

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	IRAS-16293-2422	RA: 16 32 22.7200 (248.0946667d) Dec: -24 28 34.30 (-24.47619d) Equinox: J2000	Epoch of Position: 2000	
	Comments: Pointing between A and B Category=ISM Description=[Protostars] Extended=YES				
	(2)	bkg	RA: 16 32 23.0809 (248.0961704d) Dec: -24 28 45.11 (-24.47920d) Equinox: J2000		
	Comments: Background Category=Calibration Description=[Telescope/sky background]				
	(3)	TA-star	RA: 16 32 22.1400 (248.0922500d) Dec: -24 28 9.00 (-24.46917d) Equinox: J2000		
	Comments: TA star Category=Calibration Description=[Target acquisition test] Extended=NO				

Proposal 10523 - Observation 1 - Distilling complex organics in the iconic protostellar system IRAS 16293-2422

Observation	Proposal 10523, Observation 1: science												Fri Mar 13 19:09:32 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[background (Obs 2)]												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	IRAS-16293-2422	RA: 16 32 22.7200 (248.0946667d) Dec: -24 28 34.30 (-24.47619d) Equinox: J2000				Epoch of Position: 2000						
	Comments: Pointing between A and B												
	Category=ISM												
	Description=[Protostars]												
Extended=YES													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	3 TA-star	F1000W	FAST	4	1	1	11.1	174759				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		YES			FULL			Allow Auto Reorder				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	2	2	10.0		10.0		0.0		0.0		DEFAULT		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1000W	FASTR1	5	21	1	Dither 1	4	84	1387.52	174759
	1	LONG(C)	MRSLONG		SLOWR1	15	1	1	Dither 1	4	4	1433.395	174759
	1	LONG(C)	MRSSHORT		SLOWR1	15	1	1	Dither 1	4	4	1433.395	174759
	2		IMAGER	F560W	FASTR1	31	10	1	Dither 1	4	40	3540.951	174759
	2	MEDIUM(B)	MRSLONG		SLOWR1	38	1	1	Dither 1	4	4	3631.268	174759
	2	MEDIUM(B)	MRSSHORT		SLOWR1	38	1	1	Dither 1	4	4	3631.268	174759
	3		IMAGER	F770W	FASTR1	15	8	1	Dither 1	4	32	1409.72	174759
	3	SHORT(A)	MRSLONG		SLOWR1	15	1	1	Dither 1	4	4	1433.395	174759
	3	SHORT(A)	MRSSHORT		SLOWR1	15	1	1	Dither 1	4	4	1433.395	174759

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 10523 - Observation 2 - Distilling complex organics in the iconic protostellar system IRAS 16293-2422

Observation	Proposal 10523, Observation 2: background												Fri Mar 13 19:09:32 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [science (Obs 1)]												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	bkg	RA: 16 32 23.0809 (248.0961704d)										
			Dec: -24 28 45.11 (-24.47920d)										
			Equinox: J2000										
Acquisition	Comments: Background												
	Category=Calibration												
Template	Description=[Telescope/sky background]												
	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS			YES			FULL			Allow Auto Reorder		
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	2	2	10.0		10.0		0.0		0.0		DEFAULT		
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				BACKGROUND				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1000W	FASTR1	5	21	1	Dither 1	2	42	693.76	
	1	LONG(C)	MRSLONG		SLOWR1	15	1	1	Dither 1	2	2	716.698	
	1	LONG(C)	MRSSHORT		SLOWR1	15	1	1	Dither 1	2	2	716.698	
	2		IMAGER	F560W	FASTR1	31	10	1	Dither 1	2	20	1770.476	
	2	MEDIUM(B)	MRSLONG		SLOWR1	38	1	1	Dither 1	2	2	1815.634	
	2	MEDIUM(B)	MRSSHORT		SLOWR1	38	1	1	Dither 1	2	2	1815.634	
	3		IMAGER	F770W	FASTR1	15	8	1	Dither 1	2	16	704.86	
	3	SHORT(A)	MRSLONG		SLOWR1	15	1	1	Dither 1	2	2	716.698	
	3	SHORT(A)	MRSSHORT		SLOWR1	15	1	1	Dither 1	2	2	716.698	

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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